
EE 66
Fall 2026

Signals, Dynamics, and Information

Homework 1

This homework is due Friday, September 4th, 2026, at 23:59.

Submission Format

Your homework submission should consist of a single PDF file that contains all of your answers (any hand-written answers should be scanned).

Submit the file to the appropriate assignment on Gradescope.

1. Complex Proofs (Adapted from EE 120 Homework)

Consider a pair of complex numbers v and z . Prove each of the following assertions.

- (a) $z + \bar{z} = 2\Re(z)$, where $\Re(z)$ denotes the real part of z .
- (b) Let $z = a + ib$, where $a, b \in \mathbb{R}$. Then $z\bar{z} \geq 0$, with equality if, and only if, $z = 0$.
- (c) z is real if, and only if $z = \bar{z}$.
- (d) $\overline{(zv)} = \bar{z}\bar{v}$.

2. Vector Basics

Evaluate the following expressions:

(a) In $\mathbb{R}^3$, evaluate $\mathbf{e}_1 - 3\mathbf{e}_2 - 4\mathbf{e}_3$

(b) $\mathbf{1}^\top \begin{bmatrix} a \\ b \\ c \\ d \\ e \end{bmatrix}$

(c) $\mathbf{x}^\top \mathbf{x}$, where $\mathbf{x} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$

(d) $(\mathbf{e}_1 + \mathbf{e}_2)^\top \mathbf{x}$, where $\mathbf{x} = \begin{bmatrix} 0.5i \\ 3.7 \\ 4.3 + 3i \end{bmatrix}$

(e) $\begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}^\top \begin{bmatrix} a \\ b \\ c \\ d \\ e \end{bmatrix}$

3. Inner Product Properties

Our definition of the inner product in $\mathbb{R}^n$ is:

$$\langle \mathbf{x}, \mathbf{y} \rangle = x_1 y_1 + x_2 y_2 + \dots + x_n y_n = \mathbf{x}^\top \mathbf{y}, \quad \text{for any } \mathbf{x}, \mathbf{y} \in \mathbb{R}^n$$

Prove the following identities in $\mathbb{R}^n$:

(a) $\langle \mathbf{x}, \mathbf{y} \rangle = \langle \mathbf{y}, \mathbf{x} \rangle$

(b) $\langle \mathbf{x}, \mathbf{x} \rangle = \|\mathbf{x}\|^2$

Note: $\|\cdot\|$ in this case indicates the Euclidean norm, which is defined as $\|\mathbf{x}\| = \sqrt{x_1^2 + x_2^2 + \dots + x_n^2}$

(c) $\langle -\mathbf{x}, \mathbf{y} \rangle = -\langle \mathbf{x}, \mathbf{y} \rangle$.

(d) $\langle \mathbf{x}, \mathbf{y} + \mathbf{z} \rangle = \langle \mathbf{x}, \mathbf{y} \rangle + \langle \mathbf{x}, \mathbf{z} \rangle$

(e) $\langle \mathbf{x} + \mathbf{y}, \mathbf{x} + \mathbf{y} \rangle = \langle \mathbf{x}, \mathbf{x} \rangle + 2\langle \mathbf{x}, \mathbf{y} \rangle + \langle \mathbf{y}, \mathbf{y} \rangle$

4. Subspaces Practice

For each of the sets $\mathbb{U}$ (which are *subsets* of $\mathbb{R}^3$) defined below, state whether $\mathbb{U}$ is a *subspace* of $\mathbb{R}^3$ or not.

Note:

- To show $\mathbb{U}$ is a subspace, show that it satisfies all of the necessary properties of a subspace.
- To show $\mathbb{U}$ is not a subspace, you merely have to show that at least one of the properties fails- perhaps through a counterexample.

$$(a) \mathbb{U} = \left\{ \begin{bmatrix} 2(x+y) \\ x \\ y \end{bmatrix} \mid x, y \in \mathbb{R} \right\}$$

$$(b) \mathbb{U} = \left\{ \begin{bmatrix} x \\ y \\ z+1 \end{bmatrix} \mid x, y, z \in \mathbb{R} \right\}$$

$$(c) \mathbb{U} = \left\{ \begin{bmatrix} x \\ y \\ x+1 \end{bmatrix} \mid x, y \in \mathbb{R} \right\}$$

5. Inner Products and Orthogonality

For the purposes of this question, an inner product is defined as follows:

$$\langle \mathbf{x}, \mathbf{y} \rangle = x_1 y_1 + x_2 y_2 + \dots + x_n y_n = \mathbf{x}^\top \mathbf{y}, \quad \text{for any } \mathbf{x}, \mathbf{y} \in \mathbb{R}^n$$

Show in $\mathbb{R}^2$ if two vectors are perpendicular, then they have an inner product of 0. You may **not** use the following definition: $\langle \mathbf{x}, \mathbf{y} \rangle = \|\mathbf{x}\| \|\mathbf{y}\| \cos \theta$.

Hint: The slope of a perpendicular line will be the negative reciprocal of the original line's slope. Therefore if the first vector is in the form of $\begin{bmatrix} a \\ b \end{bmatrix}$, what must the perpendicular vector look like?

6. Polynomial Spaces

- (a) Show that the set of all polynomials of the form

$$p(t) = p_0 + p_1t + p_2t^2 + p_3t^3,$$

where $t \in \mathbb{R}$ and the coefficients p_k are real scalars, forms a vector space. Call this vector space $\mathbb{V}$.

Note: Any of the coefficients p_k can be zero.

- (b) Consider the set of real-valued monomials given below:

$$\varphi_0(t) = 1, \quad \varphi_1(t) = t, \quad \varphi_2(t) = t^2, \quad \varphi_3(t) = t^3,$$

where $t \in \mathbb{R}$. Show that every real-valued polynomial of the form

$$p(t) = p_0 + p_1t + p_2t^2 + p_3t^3$$

can be written as a linear combination of the monomials $\varphi_0(t)$, $\varphi_1(t)$, $\varphi_2(t)$, and $\varphi_3(t)$. Would it be accurate to say that these constitute a basis for degree ≤ 3 polynomials? Why or why not? Recall that a basis is a set of linearly independent vectors which span a particular vector space.

- (c) Show that the set of all **derivatives** of polynomials of the form

$$p(t) = p_0 + p_1t + p_2t^2 + p_3t^3$$

forms its own vector space. Call this vector space $\mathbb{D}$.

Note: Simply recognizing that the derivatives lie in the vector space $\mathbb{V}$ from part (a) is insufficient; though if you use (and justify) this fact, it could lead you to a more-concise solution.

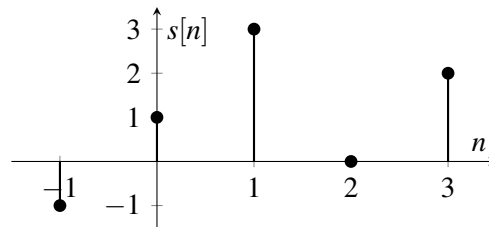
7. Signal Bases

We learned that any discrete-time signal can be written as a sum of scaled and shifted delta functions, namely

$$x[n] = \sum_{m=-\infty}^{\infty} x[m]\delta[n-m], \quad n \in \mathbb{Z}, \quad (1)$$

This is not the only useful way to rewrite a signal. In fact, there are infinitely many such "basis functions" into which a signal may be decomposed. A popular choice is the Fourier basis, which you will learn later when we cover DTFS. For now, we will work with a simpler example.

- (a) First, we will practice decomposing a signal using deltas. Write the signal $s[n]$ plotted below in the form of (1). You may assume that the signal is zero outside the pictured region.



- (b) Recall that the discrete-time unit step function is defined as

$$u[n] = \begin{cases} 0 & \text{if } n < 0, \\ 1 & \text{if } n \geq 0. \end{cases} \quad (2)$$

Write $\delta[n]$ as a linear combination of shifted unit step functions. In other words, use signals of the form $u[n-m]$.

- (c) Now write the signal $s[n]$ from part (a) in terms of shifted unit step functions. Try to do so with as few terms as possible. Then justify why any discrete-time signal can be written as a linear combination of signals of the form $u[n-m]$.
- (d) Is a decomposition of a signal into shifted unit steps unique? If your answer is "yes," explain why. Otherwise, come up with a restriction on the terms $u[n-m]$ so that the decomposition is unique. Harken back to your linear algebra class: using this conclusion and the result from part (c), what term best describes this set of unit step signals?

8. Homework Process and Study Group

Whom did you work with on this homework? List names and student ID's. (In case you met people at homework party or in office hours, you can also just describe the group.) How did you work on this homework? If you worked in your study group, explain what role each student played for the meetings this week.